

Random Matrices

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Definition 0.1. Let X be a random matrix with entries X_{ij} such that

$$X = \begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{pmatrix},$$

where $X_1, \dots, X_n$ are random vectors in $\mathbb{R}^d$ with components $X_i = (X_{i1}, \dots, X_{id})$.

Definition 0.2. The sample variance for the i -th component is given by

$$\frac{1}{n} \sum_{i=1}^n X_{i,i}^2.$$

Definition 0.3. The sample covariance between components X_{i1} and X_{i2} is given by

$$\frac{1}{n} \sum_{i=1}^n X_{i1} X_{i2}.$$

Let X be the data matrix with entries X_{ij} :

$$X = \begin{pmatrix} X_{11} & \cdots & X_{1d} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{nd} \end{pmatrix}.$$

The sample covariance matrix is defined as

$$\frac{XX^\top}{n}.$$

Definition 0.4. The eigenvalues λ_i of a matrix A satisfy

$$Av = \lambda v \quad \text{for } v \neq 0.$$

The trace of A is given by

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i.$$

Remark 0.1. The set of eigenvalues of a random matrix is called the *empirical spectral distribution*. It is defined as

$$\frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i},$$

where δ_{λ_i} is the Dirac delta function at λ_i .

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The content in the image does not explicitly contain any labeled definitions, theorems, lemmas, or proofs. It appears to be a set of notes about a random matrix with entries ± 1 with equal probability and some trace properties.

Definition 0.5. Let A be an $n \times n$ matrix with entries A_{ij} such that each A_{ij} is independently chosen to be 1 or -1 with probability $\frac{1}{2}$. The matrix A is symmetric, i.e., $A_{ji} = A_{ij}$.

Remark 0.2. For such a matrix A , the following properties hold for its eigenvalues λ_i :

$$\operatorname{tr}(A) = \sum_{i=1}^n A_{ii} = \sum_{i=1}^n \lambda_i,$$

$$\operatorname{tr}(A^2) = \sum_{i=1}^n A_{ii}^2 = \sum_{i=1}^n \lambda_i^2,$$

$$\operatorname{tr}(A^k) = \sum_{i=1}^n A_{ii}^k = \sum_{i=1}^n \lambda_i^k.$$